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A Hyperbolic Bid Stack Model for Electricity Prices – Technical Review

Overview and Modeling Approach

Context and Objectives: The paper proposes a **semi-structural price model** tailored to the Australian NEM (National Electricity Market). It extends the classic merit-order (supply stack) approach (as in Carmona et al. 2013) in **five key ways**: (1) **Generator *and* load bid curves** are modeled (introducing a "load effect" on prices), (2) **Renewable and storage (two-way) resources** are integrated structurally, (3) **Physical availability constraints** (ramping limits, minimum run levels, etc.) bound the capacity ("online fuel effect"), (4) **Network security constraints** that force or curtail dispatch are reflected via demand adjustments ("network effect"), and (5) **Price negativity and spikes** are admissible due to the functional form and regulatory price bounds [1] [2]. By capturing these features, the model aims to **trace market factors** (fuel mix changes, interconnector flows, etc.) into price outcomes while reproducing empirically observed price behaviors (e.g. occasional negative prices and extreme spikes) [3] [4]. The approach is **"semi-structural"**: it retains an economic interpretation (supply and demand curves clearing the market) but fits key components to observed bidding data rather than deriving everything from microeconomic cost assumptions [5] [6]. This relaxes the need for bids to equal marginal costs (bids can be strategic) [7]. The ultimate goal is a **detailed, replicable model** for simulating spot price paths under realistic market conditions.

Model Structure: The model is essentially a **transformation** mapping underlying inputs (demand, capacity, etc.) to the spot price each period [8] [9]. It does so by constructing an **aggregate bid stack** (supply vs demand) for a given region and solving for the market-clearing price. Key steps include: (i) building *fuel-level bid stack functions* for each category of generator and load, (ii) adjusting these curves for availability (ramping, outages, storage state) and *coalescing* them into a single merit-order stack, (iii) adjusting demand for *constrained-on/off* dispatch deviations, fixed (price-taking) bids, and inter-regional trade flows, and finally (iv) finding the price at which the adjusted inelastic demand intersects the aggregate supply curve [10] [11]. The clearing algorithm thus mirrors a security-constrained economic dispatch: low-priced generation and high-priced (willingness-to-pay) loads are dispatched first, then progressively more expensive offers until demand is met, with any "must-run" or "must-curtail" constraints treated as demand offsets [12] [13]. We now break down the model's components and equations in detail, explaining each parameter and assumption.

Fuel Bid Stack Functions (Generator vs Load)

Bid Curves Definition: The core building block is the *fuel-level bid stack function* $b_{f,i,t}(x)$, which gives the offered **price y** (in scaled units) as a function of **cumulative quantity x** (also scaled) for fuel i at time t. Separate functional forms are specified for **generator fuels** (supply-side bids, denoted $b_{G,i,t}$) and **load fuels** (demand-side bids, $b_{L,l,t}$) [5] [6]. Here a "fuel" means a category of generators or loads (e.g. coal units, gas peakers, batteries charging, etc.) aggregated for modeling purposes. The domain of x is each fuel's *available capacity range* scaled by the region's max generation capacity, so $x\in[0,1]$ in unit

terms [14] [15]. The range of y is the price in **per-unit of the regulatory price cap**, so $y\in[-1,1]$ corresponds to the allowed price floor and cap (e.g. about –$1000 and +$15,000 AUD) [16] [17]. In economic terms, $b_{G,i,t}$ is an *inverse supply curve* (price received per MW vs quantity supplied) and $b_{L,l,t}$ an *inverse demand curve* (price paid vs quantity consumed) [5] [18]. Both are *monotonic* – increasing for generators, decreasing for loads – reflecting merit-order ordering of bids [19] [20].

Hyperbolic Functional Form: The model posits that each fuel's bid curve can be well-approximated by a *shifted hyperbolic sine* function. Specifically, for generator fuel i the **generator bid stack function** is:

$$**b_{\bar{G},i,t}(x) = \alpha_t \, \sinh\Big(\beta_{i,t}\,(x - \gamma_{i,t})\Big), \tag{1}$$

and for load fuel l the **load bid stack function** is:

$$**b_{\bar{L},l,t}(x) = -\alpha_t \, \sinh\Big(\beta_{l,t}\,(x - \gamma_{l,t})\Big). \tag{2}$$

These equations appear in the paper as the defining formulas for the generator and load bid curves [21] [22]. They are **strictly monotonic hyperbolic maps** from quantity to price – $\sinh(\cdot)$ being an odd (symmetric) hyperbolic sine function. The parameter $\alpha_{t}>0$ is a **regional scale factor** common to all fuels in a region (it does *not* vary by fuel) [23]. Economically, α governs the overall **price sensitivity (elasticity) of the stack** – larger α stretches the price axis, producing steeper supply curves and thus *greater price response* to a given quantity change [23]. The parameter $\beta_{i,t}>0$ is **fuel-specific** and controls the "width" of the curve's mid-range [23]. A smaller β makes the $\sinh$ argument smaller for a given x deviation, yielding a gentler slope (more gradual price increase) over a wider quantity range; a larger β yields a sharper rise (narrower mid-range) [24]. In other words, $\beta_{i,t}$ tunes how quickly the bid price climbs from low to high as that fuel's output goes from minimum to maximum – *lower β spreads the transition over more MW (flatter curve), higher β concentrates it (steeper curve)* [23].

Finally, $\gamma_{i,t}$ (for each fuel i, including load fuels $\gamma_{l,t}$) is a **horizontal shift parameter** that is *time-varying*. It represents the **fraction of capacity offered at negative prices** [25]. Specifically, $\gamma_{i,t}$ is the *inflection point* of the curve (where $y=0$) in units of capacity [25]. If demand x is below $\gamma_{i,t}$, then for generators $b_{G,i,t}(x)$ is negative (price below \$0, implying that portion of output would be offered at a loss just to stay online), whereas if x exceeds $\gamma_{i,t}$ the price becomes positive [25]. Thus a larger γ shifts the supply curve rightward – meaning the generator is willing to supply more of its capacity at negative or very low prices before price starts rising above 0. Conversely, for loads, a larger $\gamma_{l,t}$ (in the $-\,\sinh$ formula) means the load will consume up to a higher quantity even if prices rise to 0 (or conversely, it requires prices to drop negative before taking its last increments of load). In essence, $\gamma_{i,t}$ captures **strategic bid positioning** along the quantity axis: it indicates how aggressively a fuel bids to stay dispatched (generators with high must-run tendency bid a large portion at \$-1000, etc., leading to large γ) [25]. Because market bidding behavior changes over time (e.g. more capacity bidding negative during low-demand periods, or strategic withholding in peak periods), $\gamma_{i,t}$ is modeled as a *time-dependent function* to reflect **shifting bid stacks** [26] [27]. (We detail its specification under *Calibration* below.)

Interpretation: The hyperbolic sine form ensures *symmetry* and unbounded range, which is suitable for capturing both negative prices and spikes. Importantly, the generator and load curves are symmetric in form but **opposite in slope**: one increases from negative toward positive prices (supply curve), the other

decreases from positive toward negative (demand curve). This symmetry means many properties hold analogously for supply and demand. For example, **mid-range elasticity** is governed by α for both, and the γ shift determines the zero-price crossing for both [23]. Prior models (e.g. Carmona et al. 2013) had used an exponential form for supply curves only [28]. The hyperbolic approach here generalizes that to allow *both* upward-sloping and downward-sloping curves in a unified way, accommodating the reality of **two-sided auctions** where large flexible loads (like pumped hydro or batteries charging) also submit bids [29] [30]. By fitting these continuous curves to actual bid data for each fuel, the model becomes "semi-structural": it captures the shape of aggregate offer curves without needing each unit's cost, yet remains interpretable as *implied inverse supply/demand* functions [31] [32].

Fuel Categories – Complete, Marginal, Unused: Each fuel's bid curve spans a price range from some minimum to maximum. For generators, $b_{G,i,t}(x)$ ranges from a **floor price** (often the regulatory floor if they bid some capacity at –\$1000) up to a **highest offer** (at full capacity, often near the price cap). For load fuels, $b_{L,l,t}(x)$ ranges from a **ceiling price** (highest willingness to pay, possibly near cap) down to a minimum (which could be negative if they'll be paid to consume). This allows classifying each fuel at any clearing instant as either **fully dispatched**, **partially (marginally) dispatched**, or **unused** in the solution [33] [34]. Specifically, if the market-clearing price ends up *above the entire price range* of a generator's curve, that generator fuel will be **completely dispatched** (all its available MW are used, since even its highest-price offers are cheaper than the market price) [35]. If the clearing price lies *below the entire range* of a generator's bids, that fuel is **unused** – economically it would not run at all (though if it has a non-zero minimum, it might still run as an "online" must-run unit; more on this shortly) [35] [36]. If the price falls *inside* the fuel's bid range, then the fuel is **marginal**, meaning it is partially dispatched – the market uses some fraction of its capacity up to the point where its bid price equals the clearing price [36] [37]. An analogous logic holds for load fuels but with inequalities reversed (a load is fully dispatched if clearing price is *below* its entire bid range – i.e. even its cheapest bid to buy power is willing to pay more than the market price, so it will consume its max; a load is unused if price is *above* its highest bid, meaning consuming power is too expensive; if price is in-between, the load is partially served) [34] [38]. The paper formalizes these conditions as *merit order rules*: e.g. *"the highest price in any fully-dispatched generator fuel is lower than the lowest price in any marginal generator fuel; the highest price in any marginal fuel is lower than the lowest price in any unused (or online) fuel,"* and similarly in reverse for loads [35] [34]. These inequalities ensure a **consistent merit-order sorting** of fuels by price [39]. In practical terms, **low-priced generator fuels and high-priced load fuels tend to be fully used**, whereas **high-priced generators and low-priced loads tend to be left unused** (unless physical constraints force them on). The *marginal fuels* lie in between and set the price.

Inverse Bid Functions and Market Clearing

Inverse Maps: To determine the market clearing quantitatively, we often need the *inverse* of the bid stack functions: given a price y, how many MW from a fuel would be dispatched at or below that price? The model defines inverse functions for each fuel, essentially solving $b_{f,i,t}(x)=y$ for x. Thanks to the simple hyperbolic form, this can be done in closed form. For a **generator fuel**, inverting Equation (1) yields:

$$* * b_{G,i,t}^{-1}(y) = \gamma_{i,t} \; + \; \frac{1}{\beta_{i,t}} \; \sinh^{-1}\Big(\frac{y}{\alpha_t}\Big), \tag{4}$$

where $\sinh^{-1}$ is the area hyperbolic sine (i.e. $\sinh^{-1}(z)=\ln(z+\sqrt{1+z^2})$). This gives the **quantity x (in unit capacity)** at which fuel i's offer price equals y [40] [41]. Intuitively, if the market

price is y, then generator i will be dispatched up to the point on its curve that reaches y. Similarly, for a **load fuel**, the inverse of (2) is:

$$* * b_{L,l,t}^{-1}(y) = \gamma_{l,t} \; - \; \frac{1}{\beta_{l,t}} \, \sinh^{-1}\Big(\frac{y}{\alpha_t}\Big). \tag{5}$$

Note the sign difference: because load curves slope downward, a higher price y means *less* quantity consumed (hence subtracting the term) [40] [34]. These inverses are defined on the appropriate price domains (they will return the *lowest quantity at or above price y* for generators, or *highest quantity at or above y* for loads, respecting the left- or right-continuity of dispatch) [40] [41]. In practice, the inverse functions allow **mapping a trial price to implied dispatch**: we can plug a candidate price into each $b^{-1}*{G,i,t}(y)$ to compute how many MW of each generator fuel would be supplied (all offers with bid $\le y$ would clear), and into each $b^{-1}*{L,l,t}(y)$ to get how much each load would consume (all bids with willingness $\ge y$ would be served).

Market Bid Stack (Aggregate): The *aggregate market bid stack* is conceptually the combination of all generator and load bid curves into one supply-demand diagram. Traditionally, one would sum up all individual supply curves horizontally to get a single increasing step-function of cumulative supply vs price, and likewise sum all demand curves to get a decreasing cumulative demand vs price [30] [42]. The **market clearing price** is then determined by the intersection of total supply and total demand. In this model, rather than explicit summation of step functions, the continuous hyperbolic curves and their inverses are used to replicate this process. The **market clearing condition** is that *total generation dispatched = total load served (plus net exports)* at the clearing price. Because the model deals with inelastic demand (after adjustments) on the load side, it effectively finds the price such that **aggregate supply equals a fixed demand quantity D_t** for the interval [43] [44]. Mathematically, if D_t (scaled 0–1) is the *net demand that must be met by scheduled fuels* at time t, then the clearing price in per-unit terms p_t solves:

$$* * \sum_{i \in I} b_{G,i,t}^{-1}(p_t) \; = \; \sum_{l \in L} b_{L,l,t}^{-1}(p_t) \; + \; D_t, \tag{*}$$

where I is the set of generator fuels and L the set of load fuels. Here D_t enters on the right-hand side as “demand” to be satisfied by generation (note: by construction, if there are no load-bidding fuels, D_t would equal total generation; if there are load fuels, they take some of the generation). In simpler terms, the market price is found when **the generator stack offers just enough power to meet the exogenous demand plus any power absorbed by loads bidding at that price**. The paper’s formulation of the clearing algorithm uses the *merit order categorization* described earlier: it partitions fuels into fully dispatched vs marginal vs unused based on where p_t lies relative to their bid curves [36] [45]. An **iterative algorithm** (or logical procedure) is implied:

1. **Determine Demand to Fill:** Start with the inelastic end-user demand (after adjustments, see next section) that scheduled generators must supply. This is the net demand D_t. Subtract any supply from “price-taker” generators (fixed output) and add any consumption from “price-taker” loads first, since those are dispatched regardless of price (handled as demand adjustments) [46] [47].

2. **Sort Fuels by Price:** Rank all generator fuels by their **lowest offer price** (the price at their lower capacity bound, often \$-1000 for many) and all load fuels by their **highest bid price** (at their lower bound of consumption) [33] [34]. This gives an overall merit order from cheapest generation to most expensive generation, then highest willing load down to least willing load. Conceptually, generators

on the left of the stack supply upward and loads on the right of the stack demand downward [30] [42].

3. **Initial Dispatch Allocation:** *Fully dispatch* all "cheap" generators and "expensive" loads. In practice, the model identifies **low-priced generator fuels** (those whose entire capacity is offered below higher-priced fuels) and sets them to full output, and **high-priced load fuels** (those whose entire demand curve lies above lower-priced loads) and sets them to full consumption [48] [33]. These fully-dispatched fuels are temporarily taken out of the remaining solving (their quantity is treated as fixed). Intuitively, these are "out-of-the-money" loads (willing to pay so much that they will definitely consume all they can) and "in-the-money" generators (so cheap they will run at max) given any reasonable clearing price. If any *high-priced generator* fuel has a **non-zero minimum running level** (cannot turn fully off due to technical limits), it is labeled as an **"online" fuel** and *not* entirely removed even if uneconomic – it will contribute its minimum generation regardless of price [33] [49]. This accounts for slow-start units that must stay on at a minimum generation (`online fuel effect`).

4. **Find Marginal Set:** The remaining fuels (mid-priced ones) will contain the **marginal fuel(s)** setting the price. The model partitions the price axis into segments defined by each fuel's **upper and lower bid prices** [50] [51]. Essentially, it uses each fuel's curve to mark the price at which that fuel would go from unused to marginal (its lowest bid price) and from marginal to fully used (its highest bid price). These prices create a set of contiguous price "segments." For each segment, one can determine which fuels would be fully on, which off, and which marginal if clearing price fell in that interval [50] [38]. Only one such segment will have the total supply equal to total demand. The algorithm proceeds to identify that segment and compute p_t. In practice, this is equivalent to solving equation $()$ *above, but the paper emphasizes the logical conditions:* If the market price were in segment X, then fuels A, B would be fully used, C marginal, etc., and the supply = f(p) would be Y – check against demand. *Through this piecewise solve**, the model finds the unique segment where the balance holds (by construction of the hyperbolic curves, the solution is unique and exists) [36] [37]. The price is then determined by the marginal fuel's bid function at the remaining demand to be filled. (If multiple fuels are marginal at exactly the same price – a tie – the model can handle it since the curves are continuous; in reality the dispatch engine might have tie-break rules, but here we can assume a negligible ϵ differentiates them or the iterative averaging covers it.)

5. **Price and Dispatch Outcome:** Once p_t in **scaled units** is found, it is converted to an actual price in AUD by multiplying by the price cap (since y was normalized to $P_{\max}$) [52] [53]. All fully dispatched fuels are assigned their full available MW, unused fuels get zero (or their must-run minimum if "online"), and the marginal fuel's dispatched MW is given by its inverse curve at p_t (Equations (4) or (5)). By this construction, *the least-cost dispatch matching demand is achieved*, consistent with how the real NEM dispatch engine solves a large LP (linear program) with all bids and constraints [54] [55]. The authors note that their clearing solution *coincides with a demand-based interpretation*: adjusting the demand quantity for must-run and curtailed portions (see below) yields an *"effective demand"* that directly intersects the *stack of offers sorted in merit-order* at the clearing price [13] [56]. This validates that their approach (though formulated via fuel curves and demand offsets) is equivalent to the standard market clearing logic.

In summary, the **hyperbolic bid stack** is used to clear prices by first constructing the continuous supply and demand curves for each fuel, then algorithmically determining which portions of those curves are

utilized given the (adjusted) demand. The outcome is a single regional price P_t each period such that no cheaper offers are left unused and no pricier offers or bids are unnecessarily used – mirroring an optimal merit-order dispatch. The introduction of load-bidding fuels and the handling of "online" (must-run) generation ensures that *even if some units bid above the clearing price but are forced on by technical constraints, or some loads bid below clearing but are curtailed by constraints, the model accounts for their effect by tweaking the demand input.* These tweaks are discussed next.

Demand, Trade, and Constrained-On/Off Adjustments

The model treats **demand as inelastic** in the clearing algorithm (after accounting for any flexible load bids separately). However, the *raw demand* (regional consumption) must be adjusted to account for various factors **before feeding into the market bid stack**. These include net inter-regional trade, fixed (price-taking) generation or loads, and the effects of network constraints forcing extra generation or curtailment. In effect, the model computes an **effective demand D_t** that the scheduled merit-order must meet.

Price-Taker (Fixed) Bids: In the NEM, participants can lodge *fixed volume bids* that will be dispatched regardless of price (essentially saying "I will generate X MW or consume Y MW no matter the price"). Examples are some renewable generators or certain loads with must-run contracts. The model denotes by $F^G_{i,t}$ any fixed injection by generator fuel i at time t and $F^L_{l,t}$ any fixed consumption by load fuel l [57] [58]. It also defines N^{NS}_t for *net non-scheduled generation*, i.e. aggregate output from small generators that do not participate in bidding (e.g. rooftop solar) [58]. These are summed into a **fixed net generation** term (formula (21) in the paper):

$$** F_t^{net} = N_t^{NS} \; + \sum_{i \in I} F_{i,t}^{G} \; - \sum_{l \in L} F_{l,t}^{L}, \tag{21}$$

which represents the **net supply from outside the market dispatch** [58] [59]. When determining the demand that the market must supply, this fixed net generation is *subtracted* from the region's total end-user demand. Essentially, any MWh provided by non-dispatchable or price-taker sources reduces the amount that needs to be procured through the market at time t [46]. Likewise, any fixed load (like a battery charging X MW come what may) is added to the demand that must be met. In our notation, if D^{cons}_t is the total regional consumption (including exports), and F^{net}_t as above, then the **adjusted inelastic demand** would start as $D^{cons}_t - F^{net}_t$ [46].

There is a second effect: if a *fuel* (especially a two-way fuel) has both a price-sensitive bid curve and a fixed commitment, we must ensure its dispatch accounting reflects both. The model notes that if a generator fuel i is scheduled (has $b_{G,i}$) *and* also has a non-zero fixed output $F^G_{i,t}$, then after determining the market dispatch we should **add** that fixed quantity to its dispatch result [60]. Similarly for a load fuel with fixed consumption [60]. In other words, the bid curve might only cover the flexible portion; the fixed portion is then tacked on to get the total output of that fuel. This adjustment could, in principle, slightly alter the availability for the next interval (since that fuel ran more or consumed more than the purely price-driven part), but the effect is assumed small or is absorbed into next period's capacity bounds.

Inter-Regional Trade: The NEM is divided into regions connected by transmission. The model incorporates *trade flows* by focusing on **net regional generation**. Let G^{max}_t be the region's *maximum scheduled generation capacity* (sum of all generator fuels' full capacities) and let X_t represent net exports (positive if the region exports power, negative if it imports). Then *regional consumption + exports = regional generation*.

In the paper, the demand term is said to *"incorporate trade"* by using **net generation** as the basis [61] [62]. They define an input process (often deterministic) for net generation demand: one can think of this as a forecasted baseline of how much power the region will need to produce given its own load and any export obligation. This baseline is denoted E_t (implicitly) and is modeled with a deterministic seasonal pattern (daily and yearly seasonality) – the paper gives a sinusoidal fit for it (formula (22)) [63]. Essentially, E_t captures predictable load curves (higher on hot afternoons, lower overnight, etc.) and predictable net export patterns.

Thus, combining fixed bids and net trade, the **demand that the market stack must meet** is formulated as:

$$**D_t = \frac{1}{G_t^{max}} \Big(E_t \; - \; F_t^{net} \; - \; C_t \Big), \tag{22}$$

where E_t is the baseline net generation requirement (in MW), F^{net}_t is the fixed net generation from (21), and C_t is the *constrained-on/off adjustment* (next topic) [64] [65]. This D_t is scaled by the region's max capacity G^{max}_t to express it in per-unit (0–1) of capacity [64]. In words: take the total generation needed (E_t for consumption plus exports), subtract generation that is already provided by must-run sources (F^{net}_t), subtract or add the net effects of binding constraints (C_t), and scale. This yields the *net inelastic demand* that the merit-order dispatch must satisfy. The constrained-on/off term C_t is included here because network constraints effectively make the "real demand seen by the merit order" differ from actual load – we discuss this now.

Constrained-On/Off Adjustments: In real dispatch, transmission and security constraints can force certain generators **on** (even if they are high-priced) to maintain system stability, or force certain cheap generators **off** due to congestion or security limits. Likewise, some load may be **shed (off)** despite willingness to pay, or some load **added (on)** (e.g. pumping load that must run to absorb excess generation) for system reasons. The NEM dispatch engine handles these via constraint equations, resulting in some units getting dispatch targets inconsistent with the pure price rank. The model's innovation is to capture the *price effect* of such constraints by adjusting the demand quantity input to the market stack [55] [66]. Four quantities are defined (for each region and time t) to measure the total "out-of-merit" dispatch:

- $Q^{G,on}_t$ = **Constrained-On Generation** (MW forced on). This is the total generator output in the region that is dispatched even though its bid price is *above* the market price (i.e. expensive generation that wouldn't be used based purely on price) [67]. Physically, these are units that had to run due to transmission limits or system security (e.g. a local unit must run to provide voltage support).

- $Q^{G,off}_t$ = **Constrained-Off Generation** (MW curtailed). This is the total generation in the region that is *not* dispatched even though its bid price is *below* the market price [68]. In other words, cheap supply that should have been used (price-wise) but couldn't be due to network constraints (e.g. wind farm curtailed because a line is overloaded).

- $Q^{L,on}_t$ = **Constrained-On Load** (must-run load, MW). This represents load that *is* served even though its bid price is *below* the market price [69]. That is, some loads got power despite the market price being higher than what they would normally pay – implying the system had excess generation that needed to be absorbed or a must-serve obligation (these could be reliability must-run contracts or emergency loads that were switched on to prevent shutdown of generators).

- $Q^{L,off}_t$ = **Constrained-Off Load** (curtailed load, MW). This is load that is *not* served even though its bid (willingness to pay) is *above* the market clearing price [70]. Essentially, these consumers were willing to pay the price but *could not receive energy* due to, say, local network issues (a form of load shedding or unmet demand despite price incentives).

These definitions align with how binding constraints cause deviations from an unconstrained merit order [55] [71]. Now, the **effect on price** is modeled by adjusting the demand D_t by these quantities. The logic:

- If *extra expensive generation* $Q^{G,on}$ had to run, it means the market price was artificially low (we had to force on units that normally wouldn't clear). To reflect this, the model **reduces effective demand by $Q^{G,on}$** – effectively saying, "if not for the constraint, we would have needed that much less demand for generation at the margin" [72] [73]. Reducing demand will *raise* the clearing price (since supply equals a smaller demand, we move up the supply curve to balance), mimicking the price increase that should have occurred to induce that generation economically.

- If *cheap generation was curtailed* $Q^{G,off}$, it means price was higher than it should have been (cheap supply was left out). The model handles this by **increasing effective demand by $Q^{G,off}$** [68] [74]. Intuition: to clear more demand, the price would have to fall, which is consistent with including those cheap MW (increasing demand relative to supply pushes the intersection to a lower price). Another view: we "pretend" demand was higher by the amount of lost cheap supply so that the clearing solution uses more low-priced offers, thus yielding a lower price – counteracting the constraint's impact.

- If *load was forced on* $Q^{L,on}$ (some load got served even at high price), that exerts an upward pressure on price (it absorbs excess generation). To capture it, the model **increases demand by $Q^{L,on}$** as well [69] [75]. Additional demand tends to raise the clearing price, as expected when more load must be satisfied.

- If *load was shed* $Q^{L,off}$ (some willing consumers couldn't get energy), this tends to lower the price (the market was effectively oversupplied relative to what it would otherwise serve). The model **decreases effective demand by $Q^{L,off}$** [70], i.e. we assume actual demand was that much less, which would yield a higher price intersection (since with less demand, price would not have needed to drop as much – thus preventing the price from being as low as it appeared). Decreasing demand offsets the downward price bias caused by unmet high-valued consumption.

In summary, the **net constrained quantity** C_t is defined (in formula (23)) as the sum of these adjustments:

$$**C_t = Q_t^{G,on} - Q_t^{G,off} + Q_t^{L,on} - Q_t^{L,off}, \tag{23}$$

where signs are chosen such that a positive C_t means *net demand reduction* (constrained-on generation or off-load dominating), and a negative C_t means *net demand increase* (constrained-off gen or on-load dominating). The paper states simply that C_t is "the sum of the above introduced variables" [76], implicitly with those signs as per the achieved effects. This C_t enters the demand formula (22) by being subtracted: $D_t = \frac{1}{G^{max}}(E_t - F^{net}_t - C_t)$ [64] [77]. Thus, *positive C_t (must-run gen or shed load)* reduces D_t, raising price, and *negative C_t (curtailed gen or must-run load)* increases D_t, lowering price – exactly as above logic demands.

Stochastic Treatment: The constrained-on/off adjustment C_t is not known deterministically. The authors treat C_t as a **stochastic process with mean zero** [78] [79]. Over a long sample, on average the upward and downward adjustments cancel out (hence mean zero), but at any given time there is uncertainty. They model C_t by drawing from an empirical or fitted distribution each interval, adding to the random variability in price outcomes [80] [81]. In implementation, C_t is one of the sources of randomness (along with renewable availability) that cause the model to **monte-carlo simulate multiple realizations** per time interval [82] [83]. The current implementation uses 25 iterations for each t and then averages to get the expected price path [82]. This captures the idea that we cannot predict exactly when a transmission constraint will bind and force redispatch, but we know the typical magnitude and frequency of such events – so we incorporate that risk via C_t.

Summary: After all these inputs, the **effective demand D_t** fed into the clearing engine already accounts for: (a) baseline regional consumption (with seasonality), (b) net exports/imports (via using net generation), (c) small or fixed generation and loads (subtracting F^{net}_t), and (d) anticipated network dispatch deviations (C_t). The result is a demand that when given to the merit-order stack will yield a price that *already includes the price effects of must-run units, curtailments, etc.* [13] [56]. Indeed, the authors describe C_t as a **"feasibility adjustment"** to demand that ensures the supply–demand curves "capture the true volumes after physical network constraints" [84] [85]. Without this, simply intersecting supply and demand curves on bid prices would mis-estimate price in a constrained scenario (for example, ignoring a must-run unit would undervalue price spikes). By adjusting demand first, the **market bid stack approach remains valid even under frequent constraint-induced out-of-merit dispatch** [86] [2].

Ramping, Storage, and Capacity Constraints

Another critical aspect of the methodology is how it handles **physical limits on unit availability**. In real markets, generators cannot instantly jump to max output (ramping limits), some units can't shut off below a minimum generation, and renewables are limited by weather. The model introduces **available capacity truncation** in the bid functions to reflect this [87] [88]. Rather than always using the full registered capacity of a fuel, at each time t the fuel's bid curve is *restricted* to a subset $[c_{i,t}^{lower},\;c_{i,t}^{upper}]$ of its 0–1 full range [89] [90]. This means effectively that $b_{G,i,t}(x)$ is only considered valid for x between $c_{i,t,lower}/\bar{c}*t$ and $c*t$ (where $\bar c_t$ is total max gen capacity) [89]. The model computes these }/\bar{c**availability bounds** each period from ramping and forecast considerations.

Ramp Rate Constraints: For each fuel (group of units), let $R^{up}*{i}$ and R^{down} be the aggregate ramp-up and ramp-down capability (in MW per 5-min interval, since NEM dispatch is 5-min) [91] [92]. These are determined by summing or averaging unit-level ramp rates for all units of that fuel, possibly with a safety **multiplier m** to account for the aggregation error (the model introduces m to slightly relax ramp limits knowing not all units ramp in lockstep) [93] [94]. Given the *previous interval's dispatch* of the fuel (call it $q_{i,t-1}$ in MW), the **lower availability bound** for time t is set to

$$* * c_{i,t}^{lower} = \max\{\, q_{i,t-1} - R_i^{down} \cdot \Delta t,\;\; 0 \,\}, \tag{17}$$

and the **upper availability bound** is

$$* * c_{i,t}^{upper} = \min\{\, q_{i,t-1} + R_i^{up} \cdot \Delta t,\;\; \mathrm{FullCapacity}_i \,\}, \tag{18}$$

where $\Delta t=5$ minutes here [95] [96]. Equation (17) means a fuel cannot drop below zero or more than its ramp-down amount from last period (if it was producing, it can only reduce output so fast) [97] [98]. Equation (18) means it cannot increase above last period by more than ramp-up or exceed its full registered capacity [97] [99]. These bounds are then scaled by the regional max to get the x domain for $b_{i,t}(x)$ [90] [100]. In effect, the bid curve $b_{i,t}$ is *truncated* outside these bounds – the fuel simply cannot supply more than c^{upper} or less than c^{lower} no matter the price. This directly implements the notion that *not all capacity is available if ramping from the last dispatch level is a constraint*. Notably, if a generator cannot ramp up quickly, its **upper bound will be below its full capacity** in the next interval, meaning even if price spikes, it cannot immediately reach max output (thus preventing the model from unrealistically using that capacity to cap price spikes). Likewise, ramp-down limits yield a **non-zero lower bound** $c^{lower}>0$ for certain "slow-to-start" fuels (coal, etc.), reflecting that those units **cannot turn off within one interval** [88] [101]. In the model, these fuels with persistent lower bounds are exactly the ones labeled "online fuels" when their economics would dictate shutdown [88] [101]. The model thus **anticipates minimum loading** by ensuring $c^{lower}*{i,t}$ is never below the plant's technical minimum if it's online (this is called a minimum capacity criterion, enforcing c^{lower}_i when applicable) [102] [103]. } \ge \text{MinLoad

The authors calibrate typical ramp rates from historical data for each fuel (taking care to interpret MW/min correctly for the 5-min interval) [104] [105]. They observe that slow-start baseload fuels (coal) have much tighter ramps (e.g. 5% of capacity per 5-min, so 0.05 in unit terms) than fast-start fuels (gas, hydro can be 80%+ per interval) [91] [106]. They also use a **ramp rate multiplier m** as a tuning parameter: since they aggregate multiple units into one fuel, using the sum of unit ramp rates could overstate how quickly the *entire fuel category* can respond (not all units will ramp concurrently every time). The multiplier $m\le 1$ reduces the effective ramp rates to avoid over-cutting available capacity due to mis-timing [91] [107]. A value of $m=0.8$ or similar might be used for quick-start fuels to account for coordination issues [106]. This is a heuristic to mitigate potential modeling error where an overly strict truncation from ramp rates could cause artificial "mid-price spikes" by prematurely excluding capacity that *would* actually be there a few intervals later [108] [109]. The authors note this effect and suggest using slightly relaxed ramp limits or even considering *ramp-rate-implied must-run* adjustments if a fuel is cut back too far and then spikes the price next interval when it can't ramp up in time [108] [109]. The multiplier m is one such mitigation, effectively saying the fuel can ramp a bit more in the model than a literal reading of data might suggest (to avoid pathological cases) [110] [111].

Intermittent Renewables: For wind and solar, ramping is usually not the limiting factor (they can go from 0 to max if the resource is there), but **resource availability** is key. The model sets the *lower bound for intermittent* fuels to 0 (they can always drop to zero if the sun/wind cuts out or to curtail) [112] [113]. The *upper bound* is given by **forecasts or simple models**: The paper uses a *deterministic intraday cycle* for solar (since sunrise/sunset are predictable) and a *stochastic process* for wind. Specifically, they use a **sinusoidal function** for solar generation potential:

$$c^{upper}_{\text{solar},t} = g\Big[\sin^2\Big(\frac{\pi(t_{\text{hour}}-h_0)}{H}\Big) + \sin^2\Big(\frac{\pi(t_{\text{hour}}-h_1)}{H}\Big)\Big], \tag{19}$$

which is described as the sum of two shifted sine curves over the daylight hours, scaled between 0 and some maximum g [114]. This produces a bell-shaped availability curve peaking at midday (the two terms allow a flatter peak). For wind, they propose a **scaled Brownian motion**:

$$c^{upper}_{\text{wind},t+\Delta t} = c^{upper}_{\text{wind},t} + \sigma\sqrt{\Delta t}\,\xi_t,$$

with reflection or bounds to keep it within $[0,W_{\max}]$ [115]. In the text it's given as a discretized Brownian motion with drift possibly zero and hard lower/upper limits (so it's like *$X_{t+\Delta} = X_t + \epsilon$, then truncated to [min,max]*) [115]. This captures the *random fluctuations of wind* from one interval to the next, constrained by some min and max (the max could be the installed capacity or a forecasted ceiling). The notation in (20) indicates they scale the motion so that it stays within allowed thresholds with high probability [116]. These formulas (19) and (20) generate *time-varying upper availability* $c^{upper}*{i,t}$ for solar and wind fuels. The model then truncates $b*(x)$ at those x values – effectively meaning *even if price goes to cap, solar cannot produce beyond what sunlight allows, and wind cannot exceed its stochastic availability*.

Two-Way Storage Fuels: Some fuels can act as both generator and load (e.g. battery or pumped hydro, which can charge or discharge). The model calls these **two-way fuels** and treats them specially [117] [118]. The key concept introduced is a **storage level k** (dimensionless) that balances over time. If a two-way fuel (say a battery) generates (discharges) an amount in one interval, it reduces its ability to act as a load (charge) by that amount, and vice versa. The model represents this by noting that *generator dispatch* from a two-way fuel at time t is, say, $G_{i,t}$, and the **remaining capacity for load uptake** is proportional to the **storage headroom**. The text states: *"Two-way fuels actively balance a storage level k over time. Generator dispatch is $D_{i,t}$ at time t and the remaining capacity for load uptake is k times the maximum capacity of the fuel"* [119] [22]. This suggests that k (which would vary between 0 and 1) indicates how much "charge room" is left: if $k=1$, the storage is empty (so it can take on 100% of capacity as load); if $k=0$, the storage is full (so it cannot absorb more, but can only generate). If at time t the battery discharges (generates) a lot, k will decrease (since the storage has been depleted less, meaning it filled load obligation more); if it charged, k increases (filling the storage). The model does **not explicitly constrain k within the dispatch algorithm** – instead, it treats the storage level as *implicitly determined by the sequence of dispatch decisions*. In fact, they note an *"interesting outcome"*: if you run the model without an explicit storage limit, you can **infer the implied storage level k after the fact** by looking at the net generation vs load of that fuel over time [120]. Essentially, one can reconstruct k by integrating the difference between generation and load. If k ever goes outside [0,1], it would indicate an inconsistency; the authors imply that in normal simulations it stays feasible, or that one could adjust if not. This is a clever approach to avoid adding a complex inter-temporal constraint (storage state) into the clearing algorithm: since the horizon of simulation is not optimizing over time but just stepping, they allow the battery to behave freely each step and then **verify** the trajectory of k. In principle, one could feed back the inferred k to limit next steps (e.g. if battery got full, it can't keep charging), but the paper doesn't detail that, only mentioning that it *can be inferred* if needed [120]. They likely assume the periods studied are short enough or initial k such that storage doesn't saturate hard.

In effect, **two-way fuels have both a generator bid curve and a load bid curve** in the model (they appear in both I and L sets). At any given time, they might end up generating (if price is high enough to discharge) or consuming (if price is low enough to charge), or even idle if price is mid-range. The model's merit order will naturally determine which side of the market the two-way fuel is on: if the clearing price is such that its generator offer is below price, it will generate; if price is such that its load bid is above price, it will consume; if neither, it may be partially one side or zero. By not explicitly tying the two (except through the concept of k outside the core algorithm), they allow a flexible response. But they do ensure that if, say, a battery is generating at its max, it cannot simultaneously act as load – because presumably the load side's availability would be k times capacity, and k would be zero if fully generating (since storage content is being used). The text *"remaining capacity for load uptake is k times max capacity"* [22] suggests if it generated D, perhaps $k = 1 - D/\text{max}$ (if started empty). Without more detail, we interpret that the **initial k is unrestricted** (they likely start each day or period with a moderate k), and then if a two-

way fuel is dispatched, one can deduce new k. The authors hint that you could incorporate k explicitly if doing a multi-interval optimization, but since they do sequential simulation, they treat it post-hoc. They also mention the **"ongoing two-way fuel expansion"** in NEM and that storability cannot be ignored [121], so at least their model can include these via the dual bid curves and tracking k.

Capacity Changes and Outages: Although not a primary focus, the model can also handle **unit additions or removals** by simply changing the full capacity input for a fuel at a given date [122]. For example, if a new generator comes online, G^{max}_t and that fuel's full capacity increase at that date, affecting the scaling and availability thereafter. This is straightforward since the parameters and demand scaling D_t explicitly include G^{max}_t (max scheduled generation) [63]. Planned outages or deratings could be similarly input by reducing a fuel's full capacity or using a lower c^{upper} for that period. Thus, the model can simulate scenario analyses like "what if a major plant retires" by adjusting those capacities and seeing the price effect [123] [124].

In summary, the model's handling of physical constraints ensures **feasible, realistic supply curves** each period. Generators can't magically produce beyond ramp or resource limits, and won't disappear below min-gen. Storage can arbitrage but is implicitly limited by its state. By updating the available capacity *each interval based on last interval's dispatch*, the model introduces **time-coupling**: the dispatch in one 5-min slot affects the supply potential in the next (ramp availability, etc.) [125] [90]. This contributes to realistic temporal dynamics (e.g. prices showing inertia – a sudden demand spike might cause a *sequence* of rising prices if ramping units gradually catch up). The authors emphasize that ignoring these constraints and using full capacity always could lead to spurious outcomes (they mention that using raw bid data without truncation can make supply curves look like an "X" with demand, causing anomalies) [126] [127]. Their careful incorporation of these effects is a major improvement in fidelity.

Calibration of Model Parameters

Fitting Bid Stack Curves: A crucial step is calibrating the hyperbolic bid functions (α,β,γ) to actual market data. The authors leverage published *bid stack data* from AEMO: for each 5-min interval, generators submit up to 10 price–quantity pairs (steps), which can be aggregated by fuel type [128] [129]. They construct *stacked fuel bid data* – effectively merging all bids of units of the same fuel in a region into an empirical curve – and then perform **nonlinear least squares fitting** to determine $\alpha_{t0}, \beta_{i,t0}, \gamma_{i,t}$ that best match that data [130] [131]. The notation $\alpha_{t0}, \beta_{i,t0}$ suggests the values at *initial time t_0*, or base values, since γ is time-indexed. Indeed, they mention two approaches: (A) **Formulaic Gamma Directly:** One can *directly impose a functional form for $\gamma_{i,t}$ over time* (see below) and fit α,β (and perhaps an overall level for γ) from data; or (B) **Estimate Initial and Evolve:** One can fit $\alpha_{t_0}, \beta_{i,t_0}, \gamma_{i,t_0}$ at an initial time (or initial day) by least squares, then **hold α, β fixed** and let $\gamma_{i,t}$ vary over time according to a predetermined cycle [132] [133]. The paper actually leans toward the second as a more stable approach. They note: *"An alternative would be estimating the initial gamma parameters together with initial alpha and beta using least squares. Then over time, update gamma by the cyclical change in the formulaic form while retaining its estimated level. We would have (16)."* [132] [133]. Equation (16) basically says $\gamma_{i,t} = \gamma_{i,t_0} + [\text{seasonal oscillation}]$ – i.e. start from the fitted initial value and add a seasonal component. This approach (call it **Model B**) means α and β stay at their initial fitted values for the simulation, and only γ shifts to reflect strategic bidding changes daily/seasonally.

In contrast, **Model A** might treat $\gamma_{i,t}$ itself as a fixed sinusoidal function of time (with known amplitude and phase) and perhaps re-fit α,β periodically. The authors tested *two model variations (A and B)* in their implementation [134]. In both cases, the *initial fitting* uses actual bid data from a certain date or period (they mention using data 5 days prior for fitting in tables) [135] [136]. The fitting likely minimizes the error between actual bid prices and $b_{i,t}(x)$ at several percentile points (since each fuel's actual offer is stepwise, they may fit mid-range behavior). Indeed, in their tables they list each fuel's **full capacity, ramp rates, fixed bid MW, and "fitting date, range"** [106] [137]. For example, "Black coal – Fitting range [$0,$500]" indicates they fit the coal bid curve using data primarily in the \$0–\$500/MWh price range [138]. This likely means they emphasize the mid-price portion (since coal rarely bids above \$300–\$500 except for extreme blocks) to nail down α,β for typical operation. Extremely high price spikes might not be fully captured by the hyperbolic fit, but those are handled by having a price cap in scaling.

Strategic Bid Parameter (γ) Over Time: The time-dependence of $\gamma_{i,t}$ is crucial for capturing systematic bid shifts (e.g. more aggressive bidding at night vs day, or weekdays vs weekends). The authors propose a **cyclical formula** for γ with both daily and annual seasonality [139] [140]. They define: let d be day-of-year and t_{day} be time-of-day (in say 5-min increments). They then construct a function

$$\Gamma(t) = A_0 + A_1 \sin(2\pi d/365) + B_1 \cos(2\pi d/365) + A_2 \sin(2\pi t_{\text{day}}/T) + B_2 \cos(2\pi t_{\text{day}}/T),$$

or effectively a sum of sinusoids for yearly and daily cycles [139]. In the paper this is described as *"a sinusoidal function that captures yearly seasonality through parameters and diurnal variation for time of day using..."* [141], followed by formulas (14) and (15). Equation (14) gives $\gamma_{i,t}$ for a generator fuel as a specific instance of this function (with maybe fuel-specific amplitude) [142], and (15) for a load fuel [143]. The difference between generator and load might be just phase or baseline differences, since one might expect, for example, generators to bid more capacity at negative prices when solar output is high (middle of day for solar-heavy region) vs load fuels (like pumps) might have opposite behavior. By including these harmonics, the model can represent patterns like: *every day at 1am coal units shift γ high (off-peak, they bid a lot at negative to avoid shutting down), then mid-morning γ goes lower (they don't need to give power away as demand rises), etc.* Also seasonally: *in spring (mild demand, high renewables) perhaps γ is generally higher (more negative bidding to stay on) versus in summer (high demand, units can bid higher).* The exact calibration of those parameters A_i, B_i would come from observing historical bid stacks – e.g. regressing the inferred γ from fits against time-of-day. In Model B, they said: use the fitted γ_{i,t_0} as a baseline and then add the cyclic deviation given by these sinusoids [132]. This ensures continuity at t_0 and preserves any structural bias (e.g. one fuel is generally at $\gamma=0.2$ on average, another at 0.5 due to different economics). In Model A, one might not preserve initial γ exactly but just follow the formula (14)/(15) strictly.

Alpha and Beta Stability: The regional α_t could in principle also vary with time if overall supply elasticity changes (for instance, if many units offline, supply might be steeper). However, the authors treat α_{t0} as essentially *constant* for the simulation period [23]. It is called "regional" and same for all fuels, and they did not indicate a time-varying formula for α. Thus, presumably α is fitted to an initial period and held fixed (Model B), or possibly refit infrequently if doing scenario updates. $\beta_{i,t0}$ for each fuel is likewise taken as a constant per fuel (they index it by i and initial $t0$) [21]. The subscript notation is a bit confusing (they write α_{t0}, $\beta_{i,t0}$, perhaps meaning initial values at time 0 or simply the "0th-order" parameters). In any case, these shape parameters do not follow a known function of time in the model; only γ does. This implies the **primary source of shifting bid**

curves over time is the horizontal movement (γ), while the slope/curvature (α,β) are assumed inherent to the fuel's cost structure and technology and remain fixed during the period studied. The justification is that while generators might strategize by withholding some capacity (which looks like shifting γ), their fundamental cost curve shape doesn't drastically change hourly.

Calibration Data: The authors used 2022 data from the NEM. They specifically simulate three 31-day periods (January, May, October 2022) to test the model [144]. To get input parameters, they rely on AEMO's public data: the **"Bid Move"** or dispatch data files for unit bids and **"Price, Demand, and availability"** files for actual dispatch targets and availability [145] [146]. They map each unit's DUID to a fuel category (using AEMO's mapping) and aggregate capacity by fuel [147]. Tables 5–7 in the paper list each fuel type (coal, gas, wind, etc.) by region with parameters like full capacity, ramp speeds, etc., which were estimated from this data [138] [106]. The *fitting date* appears to be a representative day (or days) used to perform the initial curve fits for each fuel. For example, NSW Black Coal was fit on a sample with price range \$0–\$500 [106]. Quick-start fuels like hydro might have been fit including higher price ranges if they often set spikes. The presence of "4-" in the fitting date column (e.g. "4-,") is unclear in snippet, but likely means Q4 of some year or 4th day. Possibly a typo, or it might indicate a subset of days each quarter used. Regardless, the calibration involved choosing a window where each fuel had sufficient bid variation to fit the hyperbola.

Strategic Bidding Representation: By altering $\gamma_{i,t}$ over time (according to the sinusoidal patterns), the model simulates *strategic capacity withholding or dumping*. For instance, generators might bid a large fraction of capacity at \$-1000 during low demand (to ensure they stay on – this is reflected by a large γ), but during peak times they might bid more capacity at higher prices (smaller γ), effectively **shifting their supply curve leftwards to raise price**. The model captures that automatically: a smaller γ at time t means the generator's curve starts rising at a lower x, meaning it withholds some capacity from cheap range, which tends to raise the clearing price (since less low-priced supply is available until price rises). This is a stylized way to include tactical bidding (without explicitly game theory) – the *magnitude and cycle of γ can be calibrated to match observed patterns of how much capacity is typically offered at very low vs high prices over time*. Thus, **"strategic bidding" in the model is an exogenous driving process** (the sinusoid plus maybe noise) that shifts the intercept of the bid curves [148]. The authors mention implementing it by "altering the intercept variable of every fuel bid curve, i.e. shifting the bid curves over time" [148]. Here "intercept" refers to γ because that determines at $x=\gamma$ the price is zero (so it intercepts the x-axis). By doing so uniformly for all units of a fuel, it's like assuming firms in each fuel category have similar strategy that moves with demand cycles. In reality, strategies might depend on prices, but here it's correlated with time of day/year which proxies demand tightness.

Summary of Calibration Strategy: In practical steps: (1) Use historical data to determine each fuel's full capacity and classify bids. (2) Choose a sample period (or multiple) to fit α,β,γ by least squares to the aggregate bid curve of each fuel. (3) Fix α (per region) and β (per fuel) for the simulation horizon. (4) Fit a seasonal function for γ for each fuel (or each category of fuel) by looking at how the fraction of capacity at negative price varies by time. Alternatively, compute empirical γ each interval from data and fit a sinusoidal regression. (5) Initialize γ_{i,t_0} to match the actual condition at start of simulation. (6) For each interval in simulation, compute $\gamma_{i,t}$ from the formula (plus possibly randomness if they added any noise – not explicitly stated, but they might keep it deterministic for cyclic variation). (7) Likewise, estimate ramp rates from data (averages per fuel) and set the ramp multiplier m to some value (from calibration or tuning). (8) Set up C_t distribution: measure historical constrained on/off quantities (AEMO provides data on out-of-merit dispatch, or one can infer from

line flows and dispatch vs bid merit) and fit a distribution (maybe normal with mean 0 or empirical distribution). The paper implies they ensure C_t has mean zero and some variance – likely determined per region and month from data [78] [81]. (9) Use historical demand and weather to fit the deterministic demand baseline E_t (seasonal load curve). Possibly use actual demand of that month as E_t for simulation input to see how well model replicates it, or an average curve. The text says *"the input process for net regional generation… can be expressed as a deterministic function with yearly and diurnal variation (22)"* [63], implying they did fit such a function rather than use actual demand traces. So they are likely generating their own demand profile.

Thus armed with these calibrated inputs, they run the simulation as described: at each 5-min step, update wind/solar availability (random for wind, deterministic for solar), update γ (cyclic), compute C_t draw (random), compute D_t from (22), then solve the market clearing. Repeat 25 iterations and average price. Compare model price distribution to actual.

Empirical Results and Distributional Fit

The model is evaluated by how well it reproduces the statistical properties of spot prices in the chosen periods. The authors focus on **distributional accuracy**, given the importance of extremes in electricity markets. They use the **Wasserstein distance (1st order)** as the metric for comparing distributions [149]. The 1-Wasserstein (also known as Earth Mover's Distance) between two distributions intuitively measures the "work" or cost to transform one distribution into the other [149]. In this context, it accounts for differences in not just mean or variance but the *entire price distribution*, giving a single number in $/MWh that indicates how far the model's price outcomes deviate from reality on average. A smaller Wasserstein distance means the model's distribution is very close to the empirical distribution of prices.

They report results like "Table 16. Wasserstein distances" [150], which compares: (a) the **distance between the model's price distribution and the actual price distribution** for each region-month, and (b) as a benchmark, the **distances among the actual distributions themselves across different sample periods** [149] [151]. The rationale: if the model's deviation is on the order of the inherent variation in prices across different months, then the model is essentially as good as the inherent uncertainty – which is considered acceptable [149] [152]. Specifically, they say *"Comparing the distributional inaccuracy of the model to the magnitude of ex ante uncertainty in the actual price data: if the distance-to-model is approximately the same or less than the distances between the empirical distributions themselves, then the model's distribution is no further from reality than one month's prices are from another's"* [149] [152]. In such cases, the model's error might be regarded as negligible for practical purposes. They highlight cases where the model's Wasserstein distance is below this benchmark and mark those as successful fits [153] [154].

Findings: The paper's results (not all excerpted here) indicate a **reasonably good fit in some scenarios and challenges in others**. For example, they note that in **Victoria (VIC)** region, the model's price distribution matched the actual quite well, achieving among the best Wasserstein metrics [155]. The intraday pattern of prices was largely captured, and the distribution of spikes and lows was similar (except some underestimation of very large spikes in certain months) [156] [157]. In **New South Wales (NSW)** and **Queensland**, the model also performed moderately, often with distance-to-model less than or equal to cross-month distances for at least some months [157] [153]. However, in **South Australia (SA)** – a smaller region with more volatile renewables – the fit was poorer, with Wasserstein distances "ballooning" in some months [158] [159]. The model tended to produce excessive volatility in SA for May and October, overshooting the actual variance (perhaps due to difficulty calibrating wind volatility or γ shifts there) [158].

Conversely, in one month (January) it may have undershot volatility due to many quick-start fuels being constrained by a narrow fit range, limiting simulated price spikes [158] [160].

Common issues identified include: - **Negative Price Frequency:** In some cases (e.g. October in certain regions), the model did not generate enough negative price instances compared to actual. The average results omitted negative prices more than actual, which the authors attribute to *"inadequate bid curve shifts"* – i.e. the γ parameter might not have been raised enough in those intervals to simulate the real extent of negative bidding [156] [161]. This suggests refining the gamma calibration for times of extreme oversupply. - **Price Spike Magnitudes:** The model sometimes failed to produce the few largest spikes observed (e.g. actual had a couple of very high prices that the model missed entirely in October). This is partly because they average 25 iterations (so very low probability spikes might get averaged out), and possibly because the availability truncation plus C_t randomness did not push the scenario into those extremes in the limited iterations [162] [163]. - **Volatility and Distribution Shape:** In some months, the model over-emphasized near-zero prices (too many prices clumped at extremely low levels) [153] [157]. This was tied to how the bid curves were shifted – if γ was a bit off, the model might keep price at \$0 for longer periods than actual, or vice versa. For example, they mention *"over-emphasis on near-nil prices can be traced back to inadequate bid curve shifts"* [153] [161]. This likely means the strategic bidding component might need a more nuanced or stochastic element to spread prices out.

Despite these issues, the authors highlight that in several cases **the model's distributional error was within the natural variability of the market** [149] [151]. For instance, for January and May in some regions, the Wasserstein distance to the model was actually *lower* than the distance between January vs May actual distributions [153] [157]. This implies the model could capture the broad distribution of prices in those months almost as well as one month resembled another in reality – a satisfying outcome for a highly nonlinear process. They also note that **intraday patterns** (the typical daily shape of prices) were "largely correct" in many cases [164] [157], meaning the model properly reflects higher prices at certain hours, the occurrence of the duck-curve dip at noon, etc., thanks to the sinusoidal γ and demand inputs.

In terms of **extremes**: the model can produce both negative prices (via the supply curves crossing below 0 when demand is low and γ high) and very high prices (when demand is near capacity and perhaps C_t removes some cheap supply, price can approach the cap). The presence of a price cap of 1.0 in scaled units naturally limits the spikes. They caution that if one wanted to simulate scarcity prices beyond the normal range, you'd need more price classes or allow price to go unconstrained (which they don't recommend as it complicates computation) [165] [86]. In NEM, price cap is enforced, so their model adhering to it is appropriate.

Overall Assessment: The hyperbolic bid-stack model, with its parameters calibrated to historical data, proved capable of capturing **key distributional features** of NEM prices: the prevalence of **zero/negative prices during oversupply**, the frequent **moderate price levels**, and the occasional **spikes**. By explicitly incorporating *renewables, storage, ramping, and network constraints*, the model better reflects the NEM's price behavior than a simpler stack model would [166] [1]. The **merit order dynamic** from Carmona et al. (2013) is preserved but enhanced: now the *"load effect"* (demand bids) can drive prices negative when there's surplus, the *"online fuel effect"* (min gen units) can suppress price dips by preventing shutdown (leading to negative prices instead of zeros, as seen), and the *"network effect"* via C_t can cause price deviations (both spikes and dips) beyond what the pure stack would do [2] [167]. These effects were qualitatively observed in their simulations.

From a replication standpoint, the paper provides **all the formulae and inputs needed**. One would gather similar bid data, fit α,β,γ, define the C_t process, and then follow the described algorithm to simulate prices. Paying attention to the *underlying assumptions*, we note: the model assumes **inelastic end-user demand** (no demand response except the fixed bids), and treats strategic behavior via exogenous γ rather than a game-theoretic equilibrium. It assumes the distribution of C_t and renewable output can be estimated independently of price (no feedback from price to availability, which is reasonable for physical constraints). These assumptions simplify the modeling greatly and seem to hold in practice for short-term spot price dynamics.

In conclusion, *"A Hyperbolic Bid-Stack for Electricity Prices"* presents a comprehensive mathematical framework that merges real-world market design elements with a tractable functional form. Each major formula in the model has a clear interpretation: **(1)-(2)** shape the supply and demand offer curves with parameters linked to elasticity and capacity; **(3)-(6)** (implied) invert those curves for solving market equilibrium; **(14)-(16)** govern how strategic bidding shifts with time; **(17)-(20)** impose physical availability limits (ramp and resource); **(21)-(22)** assemble the net demand after accounting for must-run and trade; **(23)-(24)** adjust for network constraint effects stochastically; and finally **(12)** (the market clearing solution) yields the price. The empirical calibration and tests show that when these pieces work in concert, the model can replicate complex price behaviors in a realistic power system, making it a valuable tool for scenario analysis and risk management [123] [121].

Sources: The above review is based on a detailed reading of *Katona, Nikitopoulos & Schlögl (2023)* [5] [67], including all key equations and methodological descriptions provided in that paper. The interpretations of parameters and results are drawn from the paper's own explanations [23] [25] and summary of findings [149] [156].

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[1] [2] [5] [6] [8] [9] [10] [11] [12] [13] [14] [15] [16] [17] [18] [19] [20] [22] [23] [24] [25] [26] [27] [28] [31] [32] [33] [34] [35] [36] [37] [38] [39] [40] [41] [43] [44] [45] [46] [47] [48] [49] [50] [51] [54] [55] [56] [57] [58] [59] [60] [61] [62] [63] [64] [65] [66] [67] [68] [69] [70] [71] [72] [73] [74] [75] [76] [77] [78] [79] [80] [81] [82] [83] [84] [85] [86] [87] [88] [89] [90] [91] [92] [93] [94] [95] [96] [97] [98] [99] [100] [101] [102] [103] [104] [105] [106] [107] [108] [109] [110] [111] [112] [113] [114] [115] [116] [117] [118] [119] [120] [122] [124] [125] [130] [131] [132] [133] [134] [135] [136] [137] [138] [139] [140] [141] [142] [143] [144] [145] [146] [147] [148] [149] [150] [151] [152] [153] [154] [155] [156] [157] [158] [159] [160] [161] [162] [163] [164] [165] [166] [167] A Hyperbolic Bid Stack Approach to Electricity Price Modelling
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[3] [4] [7] [21] [29] [30] [42] [52] [53] [121] [123] [126] [127] [128] [129] A Hyperbolic Bid Stack Approach to Electricity Price Modelling
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